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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services: Design and Experimental Characterization

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Master's in Telecommunications and Computer Engineering

Supervisors:

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Ph.D. Professor Pedro Ramos, Assistant Professor,
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Month, Year

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Write here your dedication.

Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

Abstract

Write here the abstract in English. There is a 250 word limit.

Keywords: *Keywords; separated; by; semicolons*

Contents

List of Figures

List of Tables

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List of Acronyms

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CHAPTER 1

Introduction

Orchestrating edge web services means dealing with coordination delays between routing, auto-scaling and monitoring components. This has been shown to negatively affect overhead on overall infrastructure and QoS due to the unaccounted effects each module has on each other. In this chapter the context, motivation and problem statement are explained in the 1.1 with the objectives presented in 1.2. The research questions that emerged are available on 1.3. The methodology used to develop the thesis is detailed in 1.4, the main contributions in 1.5 and finally the outlining of the rest of the 7 chapters of the thesis are presented in 1.6.

1.1. Context, Motivation, and Problem Statement

1.2. Objectives

1.3. Research Questions

1.4. Research Methodology

1.5. Main Contributions of this Dissertation

1.6. Dissertation Structure

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CHAPTER 2

Background and Related Work

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CHAPTER 3

Architecture and Design of Proposed Solution for Edge Services

This chapter presents the design of the proposed cross layer sdn orchestration arquitecture for edge services. It stems from the research gaps explored in the 2 and brings to the fold the need for an architecture that allows for each mentioned component to be independently managed and tested.

3.1. Design Requirements

3.2. Architecture

3.3. Distributed Elastic Allocation of Resources

3.3.1. Services

3.3.2. Data

3.4. Monitoring and Decision Engine

3.5. Control Workflow

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CHAPTER 4

Implementation of Proposed Solution

4.1. Experimental Infrastructure

4.2. Distributed Elastic Allocation of Resources

4.2.1. Services

4.2.2. Data

4.3. Monitoring and Decision Engine

4.4. Control Workflow

4.5. Implementation Validation

CHAPTER 5

Evaluation Methodology

5.1. Experimental Objectives

5.2. Evaluation Scenarios

5.3. Performance Metrics

5.4. Experimental Procedure with Results Statistical Analysis

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CHAPTER 6

Experimental Results

6.1. Trigger Quality — Overload Detection (RQ3)

6.2. Telemetry Freshness — Delivery Cadence (RQ1)

6.3. Backend Selection — Routing Awareness (RQ2)

6.4. Network Performance

6.5. Scalability Analysis

6.6. Discussion — The Compound Coordination Gap

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CHAPTER 7

Conclusions and Future Work

7.1. Conclusions

7.2. Research Contributions

7.3. Limitations

7.4. Future Research Directions